Fiscal Year:	FY 2006	**Task Last Updated:**	FY 12/23/2006
PI Name:	Sung, Chih-Jen Ph.D.		
Project Title:	CATALYZED COMBUSTION IN MICRO-PROPULSION DEVICES		
Division Name:	Physical Sciences		
Program/Discipline:	COMBUSTION SCIENCE		
Program/Discipline--Element/Subdiscipline:	COMBUSTION SCIENCE--Droplets, sprays, particles, dust clouds, nanoparticles		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	44106-7204	**Congressional District:**	11
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2001 Physical Sciences 01-OBPR-08
Start Date:	12/01/2003	**End Date:**	09/30/2006
No. of Post Docs:	2	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	3
No. of Master's Candidates:	3	**No. of Bachelor's Degrees:**	2
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA GRC
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Schneider, Steven Joseph (NASA Glenn Research Center)		
Grant/Contract No.:	NNC04AA15A		
Performance Goal No.:			
Performance Goal Text:			

The proposed program addresses the need to understand the physics and chemistry involved in propellant combustion processes in micro-scale combustors for propulsion systems and energy conversion devices on micro-spacecraft. These spacecraft are planned to have a mass less than 50 kilograms with attitude control estimated to be in the 10 milli-Newton thrust class. Combustors of this size are expected to benefit significantly from surface catalysis processes. Therefore, the proposed program specifically aims to gain a fundamental understanding of the ignition conditions, flammability limits, combustion efficiency, and modeling of propellant combustion in heated microtubes fabricated from known catalyst materials. The data and models generated will form a micro-combustor database, which is applicable to MEMS based chemical thrusters and energy conversion devices.

The present study is broken down into the following four major projects: (1) experimental determination of the ignition

Task Description:	and flammability limits as functions of Reynolds number, pressure, preheat temperature, catalyst type, and tube length to inside diameter ratio; (2) experimental mapping of wall temperature distribution and exit profiles of temperature and species; (3) experimental assessment of performance and efficiency for various micro-combustors; and (4) detailed modeling of catalytic combustion including gas phase kinetics and surface chemistry. Three different, readily available, catalyst materials, namely platinum, palladium, and nickel, will be evaluated. Gaseous propellant selections for this program include H2/O2, CH4/O2, and H2/CH4/O2. They are selected because their non-toxic character will aid in the human exploration of space. It is noted that the productions of the aforementioned fuels are near-term top priorities of in-situ resource utilization (ISRU) technologies. These proposed studies are anticipated to engender significant impacts on fundamental understanding and predictive capabilities regarding catalytic combustion in general and micro-propulsion devices using non-toxic propellants of importance to human space flight in particular. Ultimately, the studies will provide insights into approaches for achieving ignition and increased energy conversion efficiencies leading to technological advances in various combustion processes/devices in terrestrial, space-based, and extraterrestrial surroundings.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This research is directly relevant to the development of micro-satellites, which are important for Earth weather and resource reconnaissance. The present investigation can also improve the efficiency and lower the emissions of combustion devices on Earth for more efficient resource utilization with less adverse environmental impact. The micro-scale devices developed will greatly aid the portability of power supplies for field applications on Earth.
Task Progress:	A versatile test facility has been established, including a vacuum chamber, microtube reactors, mass flow control system, data acquisition system, power supply for heating the microtube, vacuum pump, and infrared camera. The vacuum chamber was outfitted with a test stand to hold the experimental setup. The test stand included ceramic insulators for mounting the microtube and a bulkhead to hold the flow connections. The chamber was also equipped with a feedthrough that includes thermocouple leads and pressure transducer wiring as well as a ZnSe window for taking IR images and several observation windows. The present study characterized the combustion of rich hydrogen/oxygen propellant mixtures and rich methane/oxygen propellant mixtures in platinum microtubes with inside diameters of 0.4 and 0.8 mm. All the mixtures tested had equivalence ratios beyond the corresponding rich flammability limits. Experimental results showed that catalytic reactions could support combustion in mixtures even when gas phase chemistry does not play a significant role. The effects of varying equivalence ratio, pressure, mass flow rate, and tube diameter on the critical temperature leading to catalytic ignition were systematically investigated. Furthermore, the effects of doping the methane/oxygen mixture with hydrogen were explored, demonstrating a substantial reduction in the ignition temperature with hydrogen addition. Microtube performance in terms of available thrust, specific impulse, and power required for preheating the microtube were also discussed. Utilizing a plug flow model, the experimental conditions were simulated with detailed gas phase/surface chemistry, thermodynamic properties, and transport properties. The computational results generally supported the experimental findings.
Bibliography Type:	Description: (Last Updated: 12/29/2006)
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